LIBXSTREAM

LIBXSTREAM is an OpenCL-based accelerator backend library providing streams, events, and device memory management for GPU offloading. It implements the DBCSR ACC interface, making it a drop-in OpenCL backend for CP2K/DBCSR.

Targets OpenCL devices across vendors (Intel, AMD, NVIDIA). Depends on LIBXS for utility infrastructure.

Build

LIBXS is taken from `LIBXSROOT` if given, otherwise from a sibling directory (`../libxs`), otherwise from where pkg-config finds it (`libxs.pc`, with `share/libxs/Makefile.inc`). An OpenCL SDK must be available.

Library — build both LIBXS and LIBXSTREAM as separate libraries:

```
git clone https://github.com/hfp/libxs.git
git clone https://github.com/hfp/libxstream.git

cd libxs && make -j $(nproc)
cd ../libxstream && make -j $(nproc)
```

This produces `lib/libxstream.a` and `lib/libxstream.so`, and with the shared library the OpenCL-MPS server `libxstream_mpsd` and its ICD `libxstream_icd.so` beside it.

Header-only (explicit) — include `libxstream_source.h` (no separate library needed for either LIBXSTREAM or LIBXS). Safe to include from multiple translation units:

```
#include <libxstream/libxstream_source.h>
```

Header-only (implicit) — compile with `-DLIBXSTREAM_SOURCE`. Any LIBXSTREAM or LIBXS public header then automatically includes the implementation. No special include order is required. `-DLIBXSTREAM_SOURCE` implies `-DLIBXS_SOURCE`; LIBXS can also be made header-only independently with `-DLIBXS_SOURCE`.

Either header-only variant needs `-D__OPENCL` (implied by `-D__OFFLOAD_OPENCL`), otherwise the implementation compiles to nothing, and the OpenCL library at link time.

The library is compiled for SSE4.2 by default but dynamically dispatches to the best ISA available at runtime (up to AVX-512). Use `SSE=0` to compile natively for the build host. The library is built into `lib/<target>`, e.g., `lib/x86_64` by default or `lib/sapphirerapids` with `SSE=0`, and `lib/` refers to the latest build. This lets a shared file system serve different systems. LIBXS is used from its matching `lib/<target>` if available.

Variable	Default	Description
GNU	1	GNU GCC-compatible compiler (0: Intel)
DBG	0	Debug build
SYM	0	Include debug symbols (-g)
SSE	1	x86 baseline: 0=native, 1=SSE4.2 (portable)
LIBARCH	(auto)	Name of the build folder (`lib/<target>`)
STATIC	(unset)	Unset: static and shared, 0: shared, 1: static
OMP	1	OpenMP (sizes per-thread resources)
OCL	2	1: `-D__OPENCL`, 2: also `-D__OFFLOAD_OPENCL`
USM	0	1: `-D__OFFLOAD_UNIFIED_MEMORY` (SVM only)

pkg-config support: `lib/pkgconfig/libxstream.pc` (`libxstream-static.pc` and `libxstream-shared.pc` select the kind). CMake: `lib/cmake/libxstream`.

Tests and samples:

```
make tests     # build and run tests/
make samples   # build every sample
```

Installation

Install into a chosen prefix (LIBXS must be built first):

```
make -j $(nproc) install PREFIX=$HOME/libxstream
```

This installs headers, the libraries, the OpenCL-MPS server (`bin/`) and its ICD (`lib/libxstream/`, versioned like the library), the pkg-config and CMake files, the sample support files (`share/libxstream/samples`), and the documentation (`share/doc/libxstream`) under `PREFIX`. The header-only source tree is optional; add `HEADER_ONLY=1` to install it. `CLEAN=1` removes an existing `PREFIX` first.

Target	Installs in addition
`install-all`	The tests (`share/libxstream/tests`)
`install-dev`	`tool_checkenvars.sh` (`PREFIX/scripts`)
`install-realall`	Both of the above, and the sample executables

Out-of-tree builds are also supported:

```
mkdir /tmp/libxstream-build && cd /tmp/libxstream-build
make -j $(nproc) -f /path/to/libxstream/Makefile
```

CMake

```
cmake -S libxstream -B build -DCMAKE_INSTALL_PREFIX=$HOME/libxstream
cmake --build build -j $(nproc) && cmake --install build
```

LIBXS is found with `find_package(libxs)` (also in `../libxs`), otherwise fetched from its repository (FetchContent), with `LIBXS_SAMPLES_OMP` set to `LIBXSTREAM_OMP`. OpenCL is required. OpenMP is required when `LIBXSTREAM_OMP` is ON. On macOS, Homebrew's `libomp` is detected via `brew --prefix libomp`; use `OpenMP_ROOT` for another installation.

Option	Default	Description
`LIBXSTREAM_OMP`	ON	OpenMP for per-thread resources
`LIBXSTREAM_INSTALL_HEADER_ONLY`	OFF	Install the header-only sources
`BUILD_TESTING`	OFF	Build the tests
`LIBXSTREAM_BUILD_SAMPLES`	`BUILD_TESTING`	Build the sample executables; requires BLAS
`BUILD_SHARED_LIBS`	OFF	Shared library, and with it the OpenCL-MPS server and ICD

A consumer uses `find_package(libxstream)` and links `libxstream::libxstream`, which also works for a tree built and installed with `make`.

Enable `LIBXSTREAM_BUILD_SAMPLES` to build the SMM, DBM, Ozaki, and stencil benchmarks under `build/samples/`. With `BUILD_TESTING`, the device-selection test also runs using the CMake-built `acc_match.x`; it needs no GPU.

API

The public C API is declared in `libxstream/libxstream.h`. All implementation details are sealed behind opaque types.

Initialization

```
int libxstream_init(void);
int libxstream_init_config(const libxstream_init_config_t* cfg);
void libxstream_init_config_default(libxstream_init_config_t* cfg);
int libxstream_finalize(void);
```

`libxstream_init()` initializes with environment/defaults. `libxstream_init_config()` allows explicit control over USM level, device selection, verbosity, and sub-buffers before the runtime starts:

```
libxstream_init_config_t cfg;
libxstream_init_config_default(&cfg);  /* -1 = use env/default */
cfg.usm = 1;      /* 0:off, 1:Intel USM, 2:SVM coarse, 3:SVM caps */
cfg.device = 0;   /* device index (-1: env/first) */
cfg.verbosity = 2;
cfg.subbuffer = 0; /* sub-buffers for offsets (no effect with USM) */
libxstream_init_config(&cfg);
```

Passing NULL to `libxstream_init_config` is equivalent to calling `libxstream_init`.

Devices

```
int libxstream_device_count(int* ndevices);
int libxstream_device_set_active(int device_id);
int libxstream_device_sync(void);
```

Streams

```
int libxstream_stream_create(libxstream_stream_t** stream_p,
                             const char* name, int flags);
int libxstream_stream_destroy(libxstream_stream_t* stream);
int libxstream_stream_sync(libxstream_stream_t* stream);
int libxstream_stream_wait_event(libxstream_stream_t* stream,
                                 libxstream_event_t* event);
int libxstream_stream_set_profiling(libxstream_stream_t* stream);
```

The `flags` are `LIBXSTREAM_STREAM_DEFAULT`, `LIBXSTREAM_STREAM_LOW`, `LIBXSTREAM_STREAM_HIGH`, or `LIBXSTREAM_STREAM_PROFILING` (`libxstream_stream_flags_t`).

Events

```
int libxstream_event_create(libxstream_event_t** event_p);
int libxstream_event_destroy(libxstream_event_t* event);
int libxstream_event_record(libxstream_event_t* event,
                            libxstream_stream_t* stream);
int libxstream_event_query(libxstream_event_t* event,
                           libxstream_bool_t* has_occurred);
int libxstream_event_sync(libxstream_event_t* event);
```

Memory

Device and host memory allocation, transfers (H2D, D2H, D2D), and initialization:

```
int libxstream_mem_allocate(void** dev_mem, size_t nbytes);
int libxstream_mem_deallocate(void* dev_mem);
int libxstream_mem_host_allocate(void** host_mem, size_t nbytes,
                                 libxstream_stream_t* stream);
int libxstream_mem_host_deallocate(void* host_mem,
                                   libxstream_stream_t* stream);
int libxstream_mem_copy_h2d(const void* host_mem, void* dev_mem,
                            size_t nbytes, libxstream_stream_t* stream);
int libxstream_mem_copy_d2h(const void* dev_mem, void* host_mem,
                            size_t nbytes, libxstream_stream_t* stream);
int libxstream_mem_copy_d2d(const void* src, void* dst,
                            size_t nbytes, libxstream_stream_t* stream);
int libxstream_mem_zero(void* dev_mem, size_t offset, size_t nbytes,
                        libxstream_stream_t* stream);
int libxstream_mem_offset(void** dev_mem, void* other, size_t offset);
int libxstream_mem_info(size_t* mem_free, size_t* mem_total);
int libxstream_mem_host_device_accessible(void);
int libxstream_mem_host_pin(void* host_mem, size_t nbytes);
int libxstream_mem_host_unpin(void* host_mem);
```

`libxstream_stream_set_profiling` enables event timestamps on a stream. `libxstream_mem_offset` yields device memory `offset` bytes past `other`. `libxstream_mem_info` reports free and total memory of the active device. `libxstream_mem_host_device_accessible` tells whether a kernel may read memory from `libxstream_mem_host_allocate` directly (requires an active device). `libxstream_mem_host_pin` declares a host range the application allocated for faster transfers, see CUDA Host Registration.

The `_hint` variants take `LIBXSTREAM_MEM_POOLED` or `LIBXSTREAM_MEM_UNPOOLED` (`libxstream_mem_pool_t`), optionally combined with a `libxstream_opencl_mem_hint_t`. Unpooled memory suits a caller that pools on its own; releasing it synchronizes the device first. Deallocation takes the hint given at allocation:

```
int libxstream_mem_dev_allocate_hint(void** dev_mem, size_t nbytes, int hint);
int libxstream_mem_dev_deallocate_hint(void* dev_mem, int hint);
int libxstream_mem_host_allocate_hint(void** host_mem, size_t nbytes, int hint);
int libxstream_mem_host_deallocate_hint(void* host_mem, int hint);
```

DBCSR Compatibility

The header `libxstream/libxstream_dbcsr.h` provides the `c_dbcsr_acc_*` symbols expected by DBCSR, allowing LIBXSTREAM to serve as a drop-in accelerator backend.

CP2K Offload Interface

The header `libxstream/libxstream_cp2k.h` implements CP2K's offload runtime interface — the `offload*` functions for general GPU operations (memory, streams, events, synchronization).

Samples

Each sample has its own Makefile under `samples/`:

```
make -C samples/smm
make -C samples/ozaki
```

SMM — Small Matrix Multiplication

Implements the ACC LIBSMM interface for batched small matrix multiply and transpose on OpenCL devices. Includes an auto-tuning framework and pre-tuned parameter sets for A100, BMG, GH200, H100, Mi250, P100, PVC, and V100. See SMM.

Ozaki — High-Precision GEMM

Ozaki scheme for high-precision GEMM emulation, fully offloaded to OpenCL. Two schemes (mantissa slicing and CRT) with automatic detection of matrix engines (Intel XMX, NVIDIA MMA). See Ozaki. The CPU-side GEMM wrapper is part of LIBXS Ozaki.

Stencil — Finite Difference

Seismic wave propagation (RTM/FWI) via dimension-split stencils. Kernel paths: FP32 (default), BF16-DPAS (`STENCIL_BF16=1`), and INT8-DPAS (`STENCIL_INT8=1`). Supports isotropic and TTI operators. See Stencil.

DBM — Distributed Block-sparse Matrices

OpenCL backend of CP2K's DBM, which processes batches of small matrix multiplications. See DBM.

License

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LIBXSTREAM Domains

CP2K Offload Interface

`libxstream/libxstream_cp2k.h` implements CP2K's offload runtime interface — the hardware-abstraction layer that CP2K uses for GPU-accelerated operations beyond DBCSR (grid integration, PW operations, etc.). The original interface provides `static inline` wrappers for CUDA, HIP, and OpenCL; LIBXSTREAM replaces the OpenCL path with a dedicated translation unit (`src/libxstream_cp2k.c`) that routes directly through LIBXSTREAM's internal API.

Relationship to the DBCSR Interface

CP2K's code has two accelerator interfaces:

Interface	Header	Purpose
DBCSR ACC	`libxstream_dbcsr.h`	Sparse matrix operations (DBCSR library)
Offload Runtime	`libxstream_cp2k.h`	General offload (memory, streams, events, synchronization)

The DBCSR adapter (`src/libxstream_dbcsr.c`) uses opaque `void*` handles and translates to LIBXSTREAM's typed API. The offload runtime adapter does the same but uses CP2K's `offloadStream_t`/`offloadEvent_t` typedefs (also `void*`). Both share the underlying LIBXSTREAM implementation.

API

The header requires only `libxstream.h` (C99, no OpenCL headers) and provides opaque handle types, an error-checking macro, and functions for error handling, streams, events, memory, transfers, and the device. The implementation is compiled only if `__OPENCL` is defined (`__OFFLOAD_OPENCL` implies it); the CMake target `libxstream::libxstream` and the pkg-config file both provide `-D__OPENCL`.

```
typedef void* offloadStream_t;
typedef void* offloadEvent_t;
typedef int   offloadError_t;

#define offloadSuccess EXIT_SUCCESS
```

```
const char* offloadGetErrorName(offloadError_t error);
offloadError_t offloadGetLastError(void);
```

`offloadGetErrorName` maps error codes to OpenCL error strings via `libxstream_opencl_strerror`. `offloadGetLastError` consumes and clears the last recorded error.

The `OFFLOAD_CHECK` macro aborts on failure after printing the error name and source location:

```
OFFLOAD_CHECK(offloadGetLastError());
```

The `offload*` functions without a return value print the OpenCL error name and code to stderr and abort when they fail.

```
void offloadStreamCreate(offloadStream_t* stream);
void offloadStreamDestroy(offloadStream_t stream);
void offloadStreamSynchronize(offloadStream_t stream);
void offloadStreamWaitEvent(offloadStream_t stream, offloadEvent_t event);
```

`offloadStreamCreate` creates a stream with default priority (`LIBXSTREAM_STREAM_DEFAULT`).

```
void offloadEventCreate(offloadEvent_t* event);
void offloadEventDestroy(offloadEvent_t event);
void offloadEventRecord(offloadEvent_t event, offloadStream_t stream);
void offloadEventSynchronize(offloadEvent_t event);
bool offloadEventQuery(offloadEvent_t event);
```

```
void offloadMalloc(void** ptr, size_t size);
void offloadFree(void* ptr);
void offloadMallocHost(void** ptr, size_t size);
void offloadFreeHost(void* ptr);
bool offloadHostMemoryDeviceAccessible(void);
```

These allocate from LIBXSTREAM's memory pools. `offloadHostMemoryDeviceAccessible` tells whether a kernel may read memory from `offloadMallocHost` directly (zero-copy). CP2K's own pool (`offload_mempool.c`) allocates through the hinted functions with `LIBXSTREAM_MALLOC_CP2K` instead (default `LIBXSTREAM_MEM_UNPOOLED`), so that its pool is not stacked on LIBXSTREAM's.

```
void offloadMemcpyAsyncHtoD(void* ptr_dev, const void* ptr_hst, size_t size, offloadStream_t stream);
void offloadMemcpyAsyncDtoH(void* ptr_hst, const void* ptr_dev, size_t size, offloadStream_t stream);
void offloadMemcpyAsyncDtoD(void* dst, const void* src, size_t size, offloadStream_t stream);
void offloadMemcpyHtoD(void* ptr_dev, const void* ptr_hst, size_t size);
void offloadMemcpyDtoH(void* ptr_hst, const void* ptr_dev, size_t size);
void offloadMemsetAsync(void* ptr, int val, size_t size, offloadStream_t stream);
void offloadMemset(void* ptr, int val, size_t size);
```

The synchronous variants (`offloadMemcpyHtoD`, `offloadMemcpyDtoH`, `offloadMemset`) pass a NULL stream. `offloadMemsetAsync` supports arbitrary fill values via `libxstream_opencl_memset`.

```
void offloadDeviceSynchronize(void);
```

```
void offloadMemcpyToSymbol(const void* symbol, const void* src, size_t count);
void offloadEnsureMallocHeapSize(size_t required_size);
```

These are CUDA-specific operations (constant-memory writes and device-heap sizing) that have no direct OpenCL equivalent. They are currently stubs guarded by assertions. CP2K's OpenCL path disables the GPU grid subsystem (`__NO_OFFLOAD_GRID`) that would call them.

See Also

- LIBXSTREAM API (`libxstream/libxstream.h`) — the underlying OpenCL backend API
- DBCSR ACC Interface — the DBCSR adapter layer
- DBM sample — OpenCL backend of CP2K's DBM
- CP2K offload_runtime.h — upstream interface definition

DBCSR ACC Interface

`libxstream/libxstream_dbcsr.h` implements the DBCSR ACC interface on top of LIBXSTREAM's OpenCL runtime — the accelerator backend contract defined by the DBCSR sparse-matrix library used in CP2K. DBCSR configured with `USE_ACCEL=opencl` uses it. DBCSR finds LIBXSTREAM with `find_package(libxstream)` (e.g., via `CMAKE_PREFIX_PATH`), links `libxstream::libxstream`, and builds LIBSMM from `samples/smm` into its own target with `libxstream_add_smm`.

Purpose

DBCSR expects every accelerator backend to expose a flat C API with the `c_dbcsr_acc_*` prefix covering six domains:

Domain	Functions	Description
Initialization	`c_dbcsr_acc_init`, `c_dbcsr_acc_finalize`	Library setup and teardown
Errors	`c_dbcsr_acc_clear_errors`	Clears the last recorded error
Devices	`c_dbcsr_acc_get_ndevices`, `c_dbcsr_acc_set_active_device`, `c_dbcsr_acc_device_synchronize`	Device enumeration, selection, and synchronization
Streams	`c_dbcsr_acc_stream_create`, `c_dbcsr_acc_stream_destroy`, `c_dbcsr_acc_stream_sync`, `c_dbcsr_acc_stream_wait_event`, `c_dbcsr_acc_stream_priority_range`	Asynchronous command queues
Events	`c_dbcsr_acc_event_create`, `c_dbcsr_acc_event_destroy`, `c_dbcsr_acc_event_record`, `c_dbcsr_acc_event_query`, `c_dbcsr_acc_event_synchronize`	Fine-grained synchronization primitives
Memory	`c_dbcsr_acc_dev_mem_allocate`, `c_dbcsr_acc_dev_mem_deallocate`, `c_dbcsr_acc_dev_mem_set_ptr`, `c_dbcsr_acc_host_mem_allocate`, `c_dbcsr_acc_host_mem_deallocate`, `c_dbcsr_acc_memcpy_h2d`, `c_dbcsr_acc_memcpy_d2h`, `c_dbcsr_acc_memcpy_d2d`, `c_dbcsr_acc_memset_zero`, `c_dbcsr_acc_dev_mem_info`	Device/host allocation, transfers (H2D, D2H, D2D), and memory queries

Every function delegates to the corresponding `libxstream_*` routine (e.g., `c_dbcsr_acc_stream_create` calls `libxstream_stream_create`). `c_dbcsr_acc_event_query` reports through `c_dbcsr_acc_bool_t` (an `int`), the header's only type.

`c_dbcsr_acc_clear_errors` acts only in OpenCL builds (`__OFFLOAD_OPENCL`). `c_dbcsr_acc_stream_create` ignores `priority` unless LIBXSTREAM is built with `LIBXSTREAM_STREAM_PRIORITIES`, which is off by default.

Profiling

The header also declares `c_dbcsr_timeset` and `c_dbcsr_timestop`, which are DBCSR's Fortran-side timer callbacks and are provided by DBCSR at link time. Every ACC function except `c_dbcsr_acc_clear_errors` is bracketed by these calls, so that individual backend operations appear in DBCSR's timing report, if two conditions hold:

- The ACC functions are compiled with `LIBXSTREAM_PROFILE_DBCSR`, which `__DBCSR_ACC` defines. A LIBXSTREAM library built on its own does not define it; this optional debug aid is enabled with `make ECFLAGS=-DLIBXSTREAM_PROFILE_DBCSR=` and such a library then only links together with DBCSR.
- `LIBXSTREAM_PROFILE` is non-zero at runtime, which also enables LIBXSTREAM's own per-kernel profile.

LIBSMM

The second half of the ACC interface, `samples/smm/acc_libsmm.h`, is not part of the library. `libxstream_add_smm` compiles it into DBCSR from `samples/smm` (see the SMM sample for its environment variables and tuned parameters):

Function	Description
`libsmm_acc_init`, `libsmm_acc_finalize`	Setup and teardown (repeated calls are allowed)
`libsmm_acc_is_thread_safe`	True if built with OpenMP
`libsmm_acc_transpose`	Batched in-place transpose of the matrices an offset stack names
`libsmm_acc_process`	Batched SMM on a homogeneous stack of `float` or `double` with M*N <= `max_kernel_dim`^{2}; other stacks return -1
`c_calculate_norms`	Not implemented (returns -1)

Utility Macros

Macro	Description
`DBCSR_STRINGIFY(SYMBOL)`	Stringifies a preprocessor token
`DBCSR_CONCATENATE(A, B)`	Concatenates two preprocessor tokens
`DBCSR_MARK_USED(x)`	Silences unused-variable warnings

See Also

- LIBXSTREAM API (`libxstream/libxstream.h`) — the underlying OpenCL backend API
- DBCSR ACC specification — upstream interface definition

OpenCL-MPS

`src/libxstream_mps.c` and `src/libxstream_mpsd.c` implement OpenCL-MPS, the multi-process service of LIBXSTREAM; its functions are declared in `libxstream/libxstream_opencl.h`. Several processes sharing one GPU are time-sliced by the device, one context at a time. OpenCL-MPS gives all processes of a node one context per device: each process forwards its OpenCL calls to a server process (`libxstream_mpsd`, installed into `bin/`), which exits once no process was attached for a while. MPI is not involved; one server runs per node and job.

Build and Installation

OpenCL-MPS is available on Linux if `CL/cl_icd.h` is found at build time, and requires the shared library (`make STATIC=0`, or CMake with `BUILD_SHARED_LIBS=ON`). The build places the server `libxstream_mpsd` and the ICD `libxstream_icd.so` beside the shared library. Installed, the server is in `bin/` and the ICD, versioned like the library, in `lib/libxstream/`; the library locates both relative to its own directory.

Usage

There are two ways to opt in:

```
libxstream_mpsd &               # a running server is the opt-in
mpirun ...

export LIBXSTREAM_MPS=1         # or: the first process starts the server
mpirun ...
```

LIBXSTREAM then points the OpenCL loader to `libxstream_icd.so` (installed into `lib/libxstream/`) when it is loaded, before any OpenCL call, which hides the installed ICDs. An OpenCL application that does not use LIBXSTREAM is served the same way, unchanged:

```
LIBXSTREAM_MPS=1 LD_PRELOAD=libxstream.so ./app
```

This relies on the library's constructor, which a C89 build (`PEDANTIC=2`) does not have. Setting both `OCL_ICD_VENDORS` (ocl-icd) and `OCL_ICD_FILENAMES` (Khronos loader) to `libxstream_icd.so` works as well. If the OpenCL driver is only found through these variables (a driver outside the default location), forwarding stays off: give their value to the server instead (`LIBXSTREAM_MPS_ICD`), and unset both.

Host buffers are shared with the server only if they come from OpenCL, i.e., with `LIBXSTREAM_XHINTS=75` (bit 4 cleared); other host memory is copied through a socket.

Limitations

The forwarding platform reports the vendor's platform and devices, with `(LIBXSTREAM-MPS)` appended to the platform name. Of several platforms, the server serves one: that of the device type `LIBXSTREAM_DEVTYPE` asks for, otherwise GPUs first. Profiling (`LIBXSTREAM_PROFILE*`), event callbacks, SVM, sub-devices, and programs from IL are not supported. A context spanning several devices is shared only with processes using the same set of devices.

Environment Variables

Variable	Description
`LIBXSTREAM_MPS`	1: forward, and start the server if none runs; 0: never forward; default: forward if a server runs
`LIBXSTREAM_MPS_TIMEOUT`	Seconds the server stays after the last process detached; negative: until killed; default: 2
`LIBXSTREAM_MPS_KEY`	Names the server to attach to; default: the job ID (`SLURM_JOB_ID`, `PBS_JOBID`, `LSB_JOBID`)
`LIBXSTREAM_MPS_SERVER`	Server executable; default: `libxstream_mpsd` in the installed `bin/` (beside the library in a build tree), else found by `PATH`
`LIBXSTREAM_MPS_ICD`	Vendor ICDs the server loads, as for `OCL_ICD_FILENAMES`; default: the installed ICDs
`LIBXSTREAM_MPS_ARENA`	Host memory in GB a process may share with the server (reserved, not committed); default: 256
`LIBXSTREAM_MPS_CONTEXTS`	Contexts per device the server spreads the processes over (round robin); default: 2 with four or more ranks per node, otherwise 1 (1 if the loader variables are set by hand)
`LIBXSTREAM_MPS_ISSUERS`	Commands the server passes to the driver at once, over all processes; default: 0 (unbounded)

Functions

Function	Description
`libxstream_mps_server`	Serve the node (what `libxstream_mpsd` runs)
`libxstream_mps_icd_function`, `libxstream_mps_icd_platforms`	ICD entry points exported by `libxstream_icd.so`

See Also

- OpenCL Backend — device selection (`LIBXSTREAM_DEVTYPE`) and the other runtime settings

OpenCL Backend

`libxstream/libxstream_opencl.h` is the internal OpenCL layer that powers every public `libxstream_*` function. It owns the OpenCL platform/device/context lifecycle, memory management, kernel compilation, and error handling. Sample code and other LIBXSTREAM extensions (e.g., LIBSMM, Ozaki) include this header to access the OpenCL runtime directly.

Compile-Time Configuration

The header is guarded by `__OPENCL` (set automatically when `__OFFLOAD_OPENCL` is defined). Key compile-time knobs:

Macro	Default	Description
`LIBXSTREAM_MAXALIGN`	2 MB	Maximum alignment for device allocations
`LIBXSTREAM_BUFFERSIZE`	8 KB	Internal scratch-buffer size
`LIBXSTREAM_MAXSTRLEN`	48	Maximum string length for names
`LIBXSTREAM_MAXNDEVS`	64	Maximum number of OpenCL devices
`LIBXSTREAM_MAXNITEMS`	1024	Per-thread maximum item count
`LIBXSTREAM_MAXNKERNELS`	32	Maximum number of distinct kernels that can be profiled
`LIBXSTREAM_PROFILE_TICKS`	10	Device-timer ticks a sample must span to be recorded
`LIBXSTREAM_USM`	1	USM support built in: 0 = none, 1 = Intel USM and OpenCL 2.0 SVM, 2 = SVM only (default with `__OFFLOAD_UNIFIED_MEMORY`); 0 without OpenCL 2.0

Runtime Configuration

Environment variables are read when the library initializes, except those concerning a device or a kernel build, which are read when a device is activated or a kernel is built. Devices are numbered in the order `LIBXSTREAM_VERBOSE` lists them.

Variable	Description
`LIBXSTREAM_VERBOSE`	0: silent, 1: errors and the activated device, 2: information (device list, build logs, warnings), 3: more; negative: everything; default: 0
`LIBXSTREAM_DEVICE`	Device to activate, by number; setting it exposes only that device (`libxstream_device_count` yields 1); default: 0
`LIBXSTREAM_DEVTYPE`	Kind of device: `gpu`, `cpu`, or `acc` (also `other`); default: all
`LIBXSTREAM_DEVIDS`	Devices to keep, by number (separated by comma, semicolon, colon, or space); default: all
`LIBXSTREAM_VENDOR`	Devices to keep, by vendor name (case-insensitive substring, e.g., `intel`); default: all
`LIBXSTREAM_DEVSPLIT`	Split each device into equal sub-devices: the number of sub-devices, or negative for one per compute unit; 1 does not split but sets `ZE_FLAT_DEVICE_HIERARCHY=COMPOSITE` (see `LIBXSTREAM_WA`); default: 0 (no split)
`LIBXSTREAM_DEVMATCH`	Device whose tuned kernel parameters are taken: a device name or ID (e.g., `[0x0bd5]`) to use another device's parameters, 0 to accept parameters of any device; default: the active device
`LIBXSTREAM_PIN`	Route for host memory, see CUDA Host Registration; default: resolved per device
`LIBXSTREAM_STAGE`	Staging window per thread in MB, for host memory staged through a runtime-owned buffer (`LIBXSTREAM_PIN=1`); default: 32
`LIBXSTREAM_STAGE_NT`	Threads of the staging copy; default: 32
`LIBXSTREAM_STAGE_GRAIN`	Block of the staging copy in KB, handed out round-robin; default: 32
`LIBXSTREAM_USM`	Unified Shared Memory level: 0 = off, 1 = Intel USM, 2 = OpenCL 2.0 SVM coarse-grain, 3 = OpenCL 2.0 SVM reported caps. Unset means 1 on an Intel discrete GPU, 0 on an Intel integrated GPU (which also transfers synchronously), off on NVIDIA, and 2 otherwise. Intel MPI cannot exchange Intel USM host memory without `I_MPI_OFFLOAD=1`: on a system with Intel GPUs only it is set unless given. If `I_MPI_OFFLOAD=0` is given, unset means 2, while an explicit 1 takes effect as requested (and fails). Where it could not be set, an explicit 1 takes host memory off Intel USM
`LIBXSTREAM_SUBBUFFER`	Sub-buffers for offset kernel-arguments (0 = off, 1 = on); unused where USM is active; default: 0
`LIBXSTREAM_CACHE`	0: no kernel cache, i.e., the vendor's compiled-kernel cache is off (`CUDA_CACHE_DISABLE=1` and `NEO_CACHE_PERSISTENT=0`, unless set); default: the vendor's cache

Level 3 is opt-in and never reached by the default; level 1 is the default on an Intel discrete GPU only, and opt-in elsewhere. Level 1 (`cl_intel_unified_shared_memory`) is the only level with asynchronous host-device transfers. Levels below 3 use coarse-grain SVM buffers only, whereas level 3 adopts whatever `CL_DEVICE_SVM_CAPABILITIES` reports (possibly fine-grain system allocations). At verbosity 2+, a warning says when an explicit level 1 cannot load the extensions, since the only symptom is a slower run.

On NVIDIA the default leaves USM off and no device memory pool is created. An explicit level is honoured there and correct, but slower: leave it unset on that vendor.

Tuning and Diagnostics

Variable	Description
`LIBXSTREAM_ASYNC`	Asynchronous operations (bitmask): 1 = H2D, 2 = D2H, 4 = D2D, 8 = memset/zero; a cleared bit makes that operation synchronous; default: 15

Variable	Description
LIBXSTREAM_XHINTS	Execution hints (bitmask): 1 = Intel thread-local queues, 2 = Intel queue families, 4 = host memory as CL_MEM_USE_HOST_PTR buffers (NVIDIA, or any vendor when the variable is set), 8 = Intel allocation flags on discrete GPUs, 16 = write-invalidate mapping of host buffers, 32 = query event status before waiting, 64 = order Intel GPUs by PCI bus, 128 = streams do not keep their last event; default: 79 (1+2+4+8+64)
LIBXSTREAM_WA	Workarounds (bitmask); default: 30, or 31 with LIBXSTREAM_DEVSPLIT=1. Unless already set: 1 sets ZE_FLAT_DEVICE_HIERARCHY=COMPOSITE, 2 EnableRecoverablePageFaults=0, 4 DirectSubmissionOverrideBlitterSupport=0, 8 UCX_RCACHE_ENABLE=0. 16 = no USM and synchronous transfers on an Intel integrated GPU, 32 = stream sync waits on a marker instead of clFinish, 64 = event sync finishes the event's queue. NEOReadDebugKeys=1 is set unless given, and NEOReadDebugKeys=0 suppresses bits 1-8
LIBXSTREAM_NLOCKS	Lock domains, 1 to 4 (main, memory, stream, event); default: 1
LIBXSTREAM_NCCS	Intel compute command streamers per device (ZEX_NUMBER_OF_CCS, unless set): 1 or 2, other non-zero: 4; default: 0 (not set)
LIBXSTREAM_INTEL	Overrides the detected Intel level: 0 = not Intel, 1 = GPU, 2 = GPU with XMX
LIBXSTREAM_NV	Overrides the detected NVIDIA level: 0 = not NVIDIA, 1 = generic, 2 = SM 7.5+, 3 = SM 8.0+, 4 = SM 9.0+
LIBXSTREAM_BIGGRF	1: Intel kernels are built with -cl-intel-256-GRF-per-thread; default: 0
LIBXSTREAM_ATOMICS	FP atomics of kernels built with libxstream_opencl_flags_atomics: non-zero number = force native, cmpxchg or xchg = emulation, 0 = unsynchronized updates (not thread-safe); default: detected
LIBXSTREAM_BARRIER	0: such kernels are built without a work-group barrier definition; default: defined
LIBXSTREAM_DUMP	1: write each kernel built from source as <name>.cl, 2 (or negative): also the program binary as <name>.dump, 3: the written source is preprocessed (see LIBXSTREAM_CPP); default: 0, or IGC_ShaderDumpEnable
LIBXSTREAM_DEBUG	Non-zero: kernels are built without -DNDEBUG (and with -gline-tables-only on an Intel GPU), build flags are printed at verbosity 1+, and DisableScratchPages=1 is set unless given; default: LIBXSTREAM_DUMP
LIBXSTREAM_CPP	1: preprocess written kernel sources, 0: never; default: with LIBXSTREAM_DUMP=3 and libxstream_opencl_dump
LIBXSTREAM_CPPBIN	Preprocessor; default: /usr/bin/cpp
LIBXSTREAM_CPPFLAGS	Extra preprocessor flags, e.g., -E -x cl for a clang driver
LIBXSTREAM_PRIORITY	Read, but without effect: stream priorities are compiled out; default: 3

Data Types

libxstream_opencl_config_t The central singleton (libxstream_opencl_config) populated by libxstream_init. It holds:

- **Device table** — ordered array of discovered cl_device_id entries.
- **Active device** (libxstream_opencl_device_t) — context, default stream, error slot, OpenCL standard level, workgroup limits, memory caps, vendor flags, and optional USM function pointers.
- **Resource pools** — lock objects, streams, events, memory-pointer registrations, and a host-memory pool (libxs_malloc_pool_t).
- **Runtime switches** — verbosity, async mode, debug/dump level, profiling, execution hints, and workaround level.
- **Histograms** — optional per-kernel duration histograms, and transfer-time histograms for H2D, D2H, D2D, and zero-fill operations.

libxstream_opencl_stream_t / libxstream_event_t Thin wrappers around cl_command_queue and cl_event respectively. Streams additionally carry a thread-ID and optional priority.

libxstream_opencl_info_memptr_t Associates a cl_mem buffer object with its host-side pointer, used to translate between SVM/USM pointers and buffer-based memory.

`libxstream_opencl_atomic_fp_t` Enumerates floating-point atomics support: none, 32-bit, or 64-bit.

Error Handling Macros

Macro	Description
`CL_CHECK(RESULT, CALL)`	Execute an OpenCL call; on failure record the error code and human-readable name
`CL_ERROR_REPORT(NAME)`	Print the last error to stderr (if verbosity is enabled)
`CL_RETURN(RESULT, NAME)`	Return from function, reporting the error if non-zero

Key Functions

Device and Context

Function	Description
`libxstream_opencl_set_active_device`	Internal device activation (lock-aware)
`libxstream_opencl_create_context`	Create an OpenCL context for a given device
`libxstream_opencl_device_name`	Return device name, platform name, and UID
`libxstream_opencl_device_level`	Query OpenCL version and device type
`libxstream_opencl_device_vendor`	Confirm a device's vendor string
`libxstream_opencl_device_ext`	Check for required OpenCL extensions
`libxstream_opencl_device_uid`	Capture or compute a unique device identifier
`libxstream_opencl_device_match`	Select the entry of a list of device names a device matches
`libxstream_opencl_info_devmem`	Query free/total/local device memory
`libxstream_opencl_device_id`	Index of a device, and optionally its global index
`libxstream_opencl_device_name_cleanup`	Strip UID brackets, decorators, and memory sizes from a device name
`libxstream_opencl_device_pageable`	Whether the active device reads pageable host memory (1, 0, or -1 if unknown)

Memory

Function	Description
`libxstream_opencl_info_devptr`	Look up a device-pointer registration (read-only)
`libxstream_opencl_info_devptr_modify`	Look up a device-pointer registration (writable)
`libxstream_opencl_info_devptr_lock`	As `libxstream_opencl_info_devptr`, with a caller-given lock
`libxstream_opencl_info_hostptr`	Look up a host-pointer registration
`libxstream_opencl_memset`	Fill device memory with an arbitrary byte pattern
`libxstream_opencl_use_cmem_size`	Whether OpenCL constant-memory hints apply
`libxstream_opencl_use_cmem`	As `libxstream_opencl_use_cmem_size` without a size
`libxstream_opencl_set_kernel_ptr`	Set a pointer kernel argument (USM-aware)

CUDA Host Registration Where the application links the CUDA runtime, host memory from `libxstream_mem_host_allocate` is registered with it (`cudaHostRegister`); otherwise, a transfer CUDA issues from such memory is pageable and slower. Nothing needs to be enabled, and there is no build or link dependency on CUDA. USM/SVM memory is registered per pool block, and blocks from plain `malloc` are not registered.

`LIBXSTREAM_PIN` selects the route for the library's own allocations and for ranges the caller declared (see below):

Value	Route
unset	Recommended: resolved once a device is activated; own allocations are registered, declared ranges are staged where the vendor runtime reports no pageable access to host memory and registered otherwise
0	Nothing is registered or staged (pageable transfers)
1	Own allocations are registered, declared ranges are staged through a runtime-owned buffer (`LIBXSTREAM_STAGE*`)
2	Own allocations and declared ranges are registered

Value	Route
3	As 2, and loads the CUDA runtime if the application did not link it, e.g., for a drop-in BLAS replacement linking only `libOpenCL`

`LIBXSTREAM_VERBOSE=2` reports how many allocations were offered and how many the CUDA runtime accepted. Memory it refused stays pageable, which shows as a slow kernel rather than as a slow copy.

A range the application allocated itself is declared with `libxstream_mem_host_pin` and withdrawn with `libxstream_mem_host_unpin`. Memory from `libxstream_mem_host_allocate` must not be declared, since it is page-locked already. Declaring neither initializes the library nor activates a device; the route applies to all declared ranges once a device is activated. Up to `LIBXSTREAM_MAXNPINS` ranges (32 by default) can be held at a time; a declaration beyond that is rejected. Failure is not fatal: the range still transfers, only slower.

Kernel Build

Function	Description
`libxstream_opencl_program`	Compile an OpenCL program from source, file, or binary
`libxstream_opencl_kernel_query`	Extract a named kernel from a compiled program
`libxstream_opencl_kernel`	Convenience: build + extract + release in one call
`libxstream_opencl_kernel_flags`	Assemble combined build flags from params, options, and extras
`libxstream_opencl_defines`	Merge user defines with internal definitions
`libxstream_opencl_flags_atomics`	Generate compiler flags for FP-atomic extensions
`libxstream_opencl_dump`	Write a kernel instance `<name>.cl` with build parameters applied, without a device
`libxstream_opencl_program_binary`	Retrieve a built program's binary (NUL-terminated)
`libxstream_opencl_retarget_ptx`	Rewrite NVIDIA PTX for the accelerated target (`sm_NNa`), optionally capping registers

Streams, Events, and Timing

Function	Description
`libxstream_opencl_stream`	Find an existing stream for a thread-ID
`libxstream_opencl_stream_default`	Return the device's default (internal) stream
`libxstream_opencl_device_synchronize`	Per-thread device synchronization
`libxstream_opencl_launch`	Launch a kernel (drop-in for `clEnqueueNDRangeKernel`, profiling-aware)
`libxstream_opencl_launch_work`	As `libxstream_opencl_launch`, with FLOP and byte counts so the profile reports rates
`libxstream_opencl_interval`	Begin and end of an event on the device clock (nanoseconds)
`libxstream_stream_priority_range`	Least and greatest stream priority (-1 if unsupported)
`libxstream_opencl_duration`	Measure elapsed seconds from a `cl_event`

Error Utilities

Function	Description
`libxstream_opencl_strerror`	Map a `cl_int` error code to a string
`libxstream_opencl_error_consume`	Clear and return the last recorded error

Profiling

Two independent facilities, reported at exit on `stderr` and silent unless requested. Both may be set at once.

Variable	Reports
`LIBXSTREAM_PROFILE`	Per-kernel durations (microseconds), one row per distinct kernel
`LIBXSTREAM_PROFILE_MEM`	Transfer rates (GB/s) for H2D, D2H, D2D, and zero-fill

A positive value sets the histogram resolution; a negative value additionally traces every sample as it is recorded, with its interval as `ns=<begin>-<end>` in device-clock nanoseconds. Setting either variable enables `CL_QUEUE_PROFILING_ENABLE` on all streams, so profiling is not meant for production runs.

Kernels are identified by `CL_KERNEL_FUNCTION_NAME`: launching through `libxstream_opencl_launch` instead of `clEnqueueNDRangeKernel` makes a kernel appear in the report.

Samples spanning fewer than `LIBXSTREAM_PROFILE_TICKS` ticks of the device timer (`CL_DEVICE_PROFILING_TIMER_RESOLUTION`) are counted as discarded rather than recorded. The report states the timer resolution and the resulting floor only when samples were dropped, and says `no samples recorded` when a requested profile collected nothing. A kernel whose every sample was discarded still gets a row that says so.

The rate on a transfer row is the histogram's mode (the most populated bucket); a kernel row reports the median duration.

Overlap A row gains one field, `inflight>=`, when its samples overlapped in time:

```
PROF ACC/OpenCL: ID=<id> ZERO=<rate> GB/s inflight>=<ratio>
PROF ACC/OpenCL: ID=<id> device inflight>=<ratio> (3 kernels, 3 transfer kinds)
```

It is a lower bound on the average number of these operations running at once (summed durations over the time their intervals cover). A value above 1 is overlap; the field is absent if no overlap shows in two decimals. Intervals arriving too late to be merged are added whole, and their count is reported.

A `device` row combines every kernel and every transfer, states how many of each contributed, and thereby shows overlap between kernels or between a kernel and a transfer.

OpenCL-MPS

Several processes can share one context per device through a server process; see OpenCL-MPS.

See Also

- LIBXSTREAM API (`libxstream/libxstream.h`) — public API built on top of this layer
- DBCSR ACC Interface — the DBCSR compatibility shim

Appendix

Presentations

- Stencils as Small Matrix Multiplications
- Practical Ozaki Schemes

Scripts

Helper scripts for building, inspecting, and maintaining LIBXSTREAM.

tool_checkabi.sh

Checks the ABI (Application Binary Interface) stability of the LIBXSTREAM shared library. The script uses `nm` to extract exported symbols from `lib/*.so` (or `lib/*.a` as fallback) and verifies that no previously published symbol has been removed or renamed compared to an earlier baseline (`.abi.txt`). Non-conforming symbol names cause an immediate error.

```
scripts/tool_checkabi.sh
```

NOTE: For full coverage the library must be built with `make STATIC=0 SYM=1` (or `DBG=1`) so that symbol information is present.

tool_checkenvars.sh

Scans the sources and public headers (`*.c` and `*.h` under `src/` and `libxstream/`) for calls to `getenv`. With `--list` it prints a sorted, deduplicated list of every environment variable used at runtime, separated into LIBXSTREAM-specific (`LIBXSTREAM_*`) and other variables. Without an argument it reports the `LIBXSTREAM_*` variables that no Markdown file documents, which is what the pre-commit hook of the same name runs; `--all` reports the deferred ones as well, i.e., those listed in `scripts/tool_checkenvars.todo`.

```
scripts/tool_checkenvars.sh --list
scripts/tool_checkenvars.sh
```

tool_checkstruct.py

Checks the structural rules of the contribution policy that span more than one line (e.g., a single function exit, blank lines). Run by the pre-commit hooks.

```
scripts/tool_checkstruct.py FILE...   # report violations, exit 1 if any
scripts/tool_checkstruct.py --list    # name the checks
```

tool_clangformat.sh

Runs `clang-format` with all arguments forwarded, preferring the newest `clang-format-N` found. `make reformat` uses it.

```
scripts/tool_clangformat.sh -style=file:.clang-format -i src/libxstream.c
```

tool_normalize.sh

Front-end for the rules in `.pre-commit-config.yaml` (requires `pre-commit`).

```
scripts/tool_normalize.sh --install   # install the Git hook (once per clone)
scripts/tool_normalize.sh [DIR]       # check and fix the tree, or just DIR
```

tool_opencl.sh

Converts OpenCL kernel files (`.cl`) — and optionally CSV parameter files — into a C header whose string literals can be compiled straight into the host binary. Include guards are stripped by default, and `#include` directives inside kernels are recursively resolved (inline).

```
scripts/tool_opencl.sh [options] infile.cl [infile2.cl ..] [infile.csv ..] outfile.h
```

Flag	Description
`-k`, `--keep`	Keep include guards (normally stripped)
`-b N`, `--banner N`	Copy the first N lines of the first `.cl` file as a banner
`-p DIR`, `--params DIR`	Directory containing CSV parameter files
`-I DIR`, `--include DIR`	Search path for angle-bracket `#include <...>` directives inside kernels (repeatable)
`-c`, `-d`, `--debug`, `--comments`	Preserve comments in the generated source

Flag	Description
`-v`, `--verbose`	Echo the command line
`-h`, `--help`	Print usage

tool_pkgversion.sh

Synchronizes the packaging metadata (RPM spec, Debian changelog) with `version.txt`, changing files only if the version differs. The Makefile runs it when it generates the version header.

```
scripts/tool_pkgversion.sh
```

tool_source.sh

Prints the header-only header (`libxstream_source.h`) that includes every library source file; an optional argument sets the directory the includes refer to (default: `../src`). `make header-only` uses it.

```
scripts/tool_source.sh >libxstream_source.h
```

tool_version.sh

Derives the project version from `version.txt` (`MAJOR.MINOR.PATCH`) and, as patch count, the number of non-merge commits since the nearest `MAJOR.MINOR.PATCH` tag (or all commits if there is none). It can emit a plain version string, individual version components, or a complete C header with `#define` guards.

```
## full version string (branch-version-patch)
scripts/tool_version.sh

## generate a C version header with a given symbol prefix
scripts/tool_version.sh LIBXSTREAM -1

## single component: 1=major, 2=minor, 3=update, 4+=patch
scripts/tool_version.sh LIBXSTREAM 2
```

Argument	Description
`PREFIX`	Symbol prefix used in the generated header (uppercased)
`CMPNT`	`0` (default) — version string; negative — C header; `1`–`3` — major/minor/update; `>3` — patch count
`SHIFT`	Added to the patch count (default: `0`)

Contributing

This is the contribution policy for LIBXS and LIBXSTREAM. Both projects follow the same policy, so the text is project-neutral: it is maintained in LIBXS (`documentation/libxs_contrib.md`) and copied into dependent projects by `make documentation`, the same way `Makefile.inc` is shared. Please edit it in LIBXS — the copy is overwritten whenever the original changes.

The code base is small, long-lived, and read far more often than it is written, so uniformity is what keeps diffs reviewable. When a rule below does not answer your case, the strongest rule is: **match the surrounding code**.

Not every rule is derived from first principles. Some are taste, some are habit, and some are plain superstition. They are the policy regardless: a uniform code base is worth more than an individually optimal choice, and rules that are merely arbitrary are still cheap to follow.

Character Encoding and Whitespace

Files	Encoding
Source, headers, kernels, Makefiles, scripts, YAML, plain text	US-ASCII only
Markdown (`*.md`)	any UTF-8

No Unicode in code or build files — no typographic quotes, no em dashes, no non-breaking spaces. Such characters survive editors, diffs, and generators poorly, and neither the compiler nor `sed` is required to handle them.

Markdown has no such restriction: a spaced em dash (—) instead of `--`, an en dash for numeric ranges (1–4), and mathematical notation (alpha, <=, ceil(n/2), x, sum) are all welcome wherever they read better than an ASCII transliteration.

Beyond encoding:

- **No tabs**, except where a Makefile requires them.
- LF line endings. CRLF is rejected.
- No trailing whitespace.
- No whitespace before `#` in a preprocessor directive.
- **No French spacing** in either sense: no space before `,`, `;`, `:`, `!`, or `?`, and a single space after a sentence-ending period. This holds for code, comments, commit messages, and documentation alike.
- A script carrying a shebang is executable in the index (mode 755); every other file is 644.

These are enforced mechanically rather than by review. `.pre-commit-config.yaml` combines the standard pre-commit hooks (trailing whitespace, line endings, byte-order marks, shebang and exec-bit consistency, YAML syntax) with the project-specific rules (US-ASCII, tabs, C++ comments, whitespace before `#`, `exit()` in library code, `sed -i` in scripts, `goto`, a declaration in a `for` initializer, banner comments, `--` in a comment, three blank lines, and column 73 in fixed-form Fortran). Install the Git hook once per clone; the same configuration runs in continuous integration, so a violation fails the pull request:

```
scripts/tool_normalize.sh --install   # once per clone
scripts/tool_normalize.sh             # check and fix the whole tree
scripts/tool_normalize.sh src         # or one directory
```

`.editorconfig` keeps most of it from happening in the first place. Data files (`*.csv`, `*.tsv`, `*.dat`) and generated sources are exempt — tabs and line endings are part of their format, and their producer owns them. Of the rules above only the two spacing conventions are checked partially: a space before a comma or semicolon is caught in C sources, the rest is on the author.

The rules that span more than one line are checked by `scripts/tool_checkstruct.py`: a single function exit, blank lines inside and between functions, stacked single-line comments, the shape of a multi-line comment, where an opening brace goes, whether a multi-line `if` or loop is braced, whether the sections of a file come in order, and a constant on the left-hand side. It works on the source text, with comments, literals, and preprocessor directives masked out, so a `return` in `#if` and another in `#else` count as one exit. A parser is not used on purpose: the declarations carry `LIBXS_API` and friends, which a preprocessor-less parser turns into an error node every few lines, and a preprocessed compiler dump no longer says which file a construct came from. `scripts/tool_checkenvars.sh` compares the prefixed variables the source reads against what the documentation mentions.

The policy holds everywhere; the hooks enforce most of it on library code and public headers. A sample or a test that bails out of `main()` early is not the defect a multi-exit library function is, so `single-exit`, `function-gap` and the rest are scoped to `src`, the public headers and the kernels. Four of them are not, because they mislead whoever reads the code wherever it sits: a comment that has come loose from what it documents, a comment whose lines do not carry the `*` that marks them, two closing braces sharing a column while their blocks are nested, and a second blank line inside a function body. Those run over the whole tree, samples and tests included: a contribution is held to the policy wherever it lands, and what that surfaces is listed rather than waived.

Both keep their open findings in a to-do file beside them, `scripts/tool_checkstruct.todo` and `scripts/tool_checkenvars.todo`, rather than in a file-level exclusion. Three properties follow, and each is the point:

- The list is **per rule and per file**, so a file listed for one rule is still checked by the others.
- The list is **per project and not propagated**. The two scripts are policy files that `make documentation` copies from LIBXS; a count lowered in a copy would be reverted, so the state cannot live inside them.

- The list is a **to-do, not a permission**. Each entry carries what it defers, and going the other way fails as well: one finding more than listed is a regression, one less means the entry outlived its findings, and an entry naming a file that no longer exists is reported too. An exclusion that outlives its cause is how a list starts lying.

The list maintains itself in the one direction that is safe. A fix followed by `tool_normalize.sh` lowers the count, and drops the entry when nothing is left; an entry whose file is gone drops too. The run still fails and says what it changed, exactly as the whitespace hooks do, so the smaller list is reviewed and committed rather than applied silently. Upwards never happens: a count that grew, or a new undocumented variable, is a regression and stays an error. `tool_checkstruct.py --counts` prints the table from scratch if a list needs rebuilding wholesale.

C Source File Structure

A translation unit is strictly grouped, in this order:

1. Includes
2. Macros
3. Types (typedefs, struct definitions)
4. Translation-unit variables (file-scope statics and globals)
5. Prototypes, where forward declarations are needed
6. Functions

No interleaving. A macro used only by the last function still belongs in the macro section, and a type or a table used only by the last function belongs in its own. This makes the shape of every file predictable: what it depends on, what it configures, what state it holds, and what it does — in that order. Grouping is also what keeps the sections reviewable: a macro parked next to its first use hides how many of them the file already has.

Three exceptions, and no others:

- A translation unit split into implementation fragments includes those fragments where their code belongs, not at the top. Such an include is a piece of the implementation rather than a dependency, and hoisting it would reorder the definitions it contains. This one is detected rather than judged, by the one property that separates the two: a fragment carries no include guard, because two of them in one translation unit would define everything twice. A dependency is quoted as well where it is a sibling, so the quotes decide nothing.
- A type may sit immediately above the one entry point it parameterizes, when that is what makes the interface readable. This covers an argument or callback type in a public header, not an internal type shared by several functions.
- In a header the last two sections are one. A header is read as a list of declarations, and an inline definition sits in that list: `LIBXS_API_INLINE` interleaved with `LIBXS_API` is the feature group the reader wants, not an interleaving to fix. The reason is the same one that lets a header open a function body on the right. It applies to those two sections only: a macro, a type or a file-scope variable below either of them is still out of order.

A macro that is undefined again is local to the code between the two, the way a variable is local to a block, so it is not a member of the macro section either. `LIBXS_FPRINT_LOAD` in `libxs_math_fprint.h` is defined above the one function that expands it and undefined below, and moving it into the macro section would separate it from its only use and leave the `#undef` reading as noise. The `#undef` has to follow the definition to mean this: the other order is the defensive one, dropping what a platform header may have set, and those macros (`__STRICT_ANSI__`, `_WIN32_WINNT`) are ordinary members of the section.

What sits inside a conditional is a feature test, and a declaration is no exception. A platform header may not declare what it should, so the declaration goes next to the `#define` that calls it and inside the same condition; a type the platform does not provide is defined next to the `#define` that records its availability. `LIBXS_EXTERN void flockfile(FILE*)` in `libxs_sync.h` and the `__int128` typedef in `libxs_macros.h` are both this, and hoisting either one means writing its condition a second time. So inside an `#if`, only a definition is ranked.

That has a prerequisite: a conditional that **wraps the whole file** must not count, or a translation unit that puts `#if defined(__OPENCL)` around everything it contains would have no rules left inside. The test for it is symmetric, since a header ends with a conditional include of its own implementation, which also has nothing after it: nothing that belongs to a section may precede the opener either, and the include guard is the same shape.

The `section-order` check reads the file as a sequence of top-level constructs and reports where their kind steps back to a section the file has already left. It is approximate by design: it ranks what it recognizes and stays quiet about

the rest, so it misses cases rather than inventing them. A construct that is nothing but a macro invocation is left unranked, since what it expands to is not in the text, and so are the statements of an included body fragment, whose control flow sits at brace depth zero without being a definition. One backward step is reported once and not once per construct below it, because a single misplaced typedef does not make every macro under it a separate defect.

The shapes ranked out of the ordering are the ones where the text that looks like a section member is not one: the include guard, whose `#define` is the file's own name; anything but a definition inside an `#if`; a prototype immediately above the definition it repeats, which is how a static definition answers `-Wmissing-prototypes`; an include of a guardless implementation fragment; and a macro that is undefined again.

What is left over is judged, one file at a time, rather than guessed at by a rule reading intent — a type above the one entry point it parameterizes has no signal a checker could read. The judgement is recorded where it stays visible: in the to-do list while the file has structure the rule must keep watching, and in the script's permanent list only where the accepted shape is the one thing that rule could report in that file. `libxs_macros.h` is the second kind, since its standard includes must follow the feature-test macros that configure them, and macros and those includes are all it holds.

Every file carries the SPDX license header (BSD-3-Clause) verbatim as found in existing files.

C Dialect

C89 (ANSI C) in general. The pre-submit configuration (`PEDANTIC=2`) compiles with `-std=c89`, so a C99-only construct breaks the build for everyone else even when it compiles for you:

- Declarations at the beginning of a block, before any statement. No declarations mixed into the middle of a block, and no declaration in a `for` initializer.
- `/* ... */` comments only.
- No variable-length arrays, compound literals, designated initializers, `restrict`, or `//`-style line continuation tricks.
- Newer facilities are reached through the macros and typedefs the public headers already provide, not by raising the dialect locally.

Where a C99 (or later) construct is genuinely required, it is guarded and confined, in the same style as the existing guards.

OpenCL kernels (`*.cl`) follow every rule above except the dialect. They are compiled as OpenCL C, which is C99, so a declaration in a `for` initializer is correct there and the hooks exempt kernels from that one rule. Everything else holds unchanged: US-ASCII, no tabs, `/* ... */` comments only, a single function exit, two blank lines between functions, capitalized macros, and the SPDX header. A kernel is source, not data.

A 64-bit integer is `uint64_t` or `int64_t`. They are exactly 64 bits wide rather than merely at least that wide, they add no name of ours to the API, and they cost the user nothing: `libxs_macros.h` already includes `<stdint.h>` and `<inttypes.h>`, so every consumer of the public headers has them, and the API already returns `uint64_t` from `libxs_hilbert` and `libxs_morton`. `long long` is kept only where something outside the project spells it — the Intel intrinsics take `long long*` (`_mm256_i64gather_epi64`), the `__atomic_*_8` builtins take `long long`, and a value printed with `%llu` is cast to `unsigned long long`, which is portable without composing the format string out of `PRIu64`. Those are the reasons `-Wno-long-long` is set for a pedantic build, and they are the only ones: new code does not reach for `long long` to hold a number of its own.

Functions

- **A function has a single exit.** No early `return`, no multiple return paths, no `goto`.
- **No trailing underscore on a parameter or a local**, in a definition or a declaration. That mark is reserved for a variable a *macro* declares, so that such a name cannot collide with one the caller already has in scope; a function wearing it anywhere takes the mark away from the one thing it is for. Passing an underscored name *to* a function is a different matter and fine: inside a macro body that is exactly what the macro's own local is for.
- Use a `result` variable, gate subsequent work on `EXIT_SUCCESS == result`, and return `result` at the single exit point.
- Constants go on the left-hand side of a comparison (`EXIT_SUCCESS == result`, `NULL != ptr`), which turns an accidental assignment into a compile error.
- Two blank lines between function definitions.
- At most one blank line inside a function body.

```
int example(const void* input, void** output)
{
  int result = EXIT_SUCCESS;
  if (NULL == input || NULL == output) result = EXIT_FAILURE;
  if (EXIT_SUCCESS == result) {
    result = prepare(input);
  }
  if (EXIT_SUCCESS == result) {
    result = finish(input, output);
  }
  return result;
}
```

Macros

A macro is capitalized, and so are its parameters: `LIBXS_ALIGN(POINTER, ALIGNMENT)`, not `LIBXS_ALIGN(pointer, alignment)`. The parameters are the part that is easy to forget, and they are the part that matters at the point of use: a capitalized argument is what tells the reader that the expression may be evaluated more than once. **A parameter carries no trailing underscore**: that marks a variable the macro declares itself, and the two must stay apart, since the whole point of the underscore is to say "this name is mine, not yours".

A variable a macro declares is the other way round: lowercase, with a trailing underscore, and ideally prefixed by the macro's own name.

```
#define MACRO() { int macro_i_ = 0; }
```

The capitalization tells the reader which names come from the call site and which the macro invented; the trailing underscore and the prefix are what keep the invented one from colliding with a variable the caller already has in scope. A macro that declares plain `i`, `s`, or `p` is a trap for whoever expands it next to their own `i`. The members of an aggregate the macro declares are not locals — in `union { float v; float a[8]; } u_` it is `u_` that carries the underscore — and a `*_DECL(A)` macro that declares a variable named by its own parameter keeps the caller's spelling.

Two kinds of macro name are lowercase on purpose, and both are exempt:

- A trailing lowercase segment is a token pasted onto the name — a type (`LIBXS_TYPECHAR_double`), a lock kind (`LIBXS_LOCK_ACQUIRE_spin`), an address space. It has to match the spelling of what it names.
- A macro standing in for something that is not ours keeps that spelling: a compiler builtin (`__builtin_nan`), a language keyword the OpenCL-on-CPU path defines away (`kernel`, `restrict`), a foreign API (`offloadSuccess`), or the lowercase alias a paste target needs (`libxs_crc32_b8`).

Comments

- **A comment adds value on top of the code.** Never document what the code obviously does — the code already says that. Document what it cannot say: why this way, what breaks otherwise, which assumption is being relied on.
- Prefer no comment at all. Then prefer one line.
- **Never stack comments.** Two comments with no code between them are either one comment — make it one line, or a block if it earns the size — or they describe different subjects, and then the code each one describes belongs between them. **A blank line between them does not separate them**: it is still two comments and no code. This covers blocks as much as single lines; the license header, being the file's leading comment, is exempt.
- **Single-line comments by default.** A multi-line comment has to earn its size: it is reserved for what is absolutely necessary, i.e., a risk or a trap worth spelling out — typically something that has already been gotten wrong once and would be gotten wrong again without the warning. Everything else is one line or nothing.
- API documentation in public headers is the other place a multi-line block belongs.
- **A multi-line block carries a `*` on every line**, under the one in the opening `/*`, the closing line included. Two shapes satisfy that and both are accepted outside a header: the block that opens on its own line, and the compact one whose text starts on the opening line and ends on the closing one. What it rules out is a third shape, a hanging indent with no stars, where nothing marks the lines as comment and a reader scanning the left edge loses them.

- **In a header, a block at file scope opens and closes on its own line.** The reason is the API rather than the comment, and it is the one that lets a header open a function body on the right: the declarations are read as a list, and a block starting and ending mid-line breaks the rhythm of that list. Inside a function body, and in an implementation unit, either shape is fine. Measured before requiring it: 452 blocks at header file scope already open on their own line and 15 do not.
- `/* ... */` only. **C++ comments (`//`) are rejected** in `.c` and `.h`.
- **No decorative or banner-style comment blocks.** No `/*==== Section ====*/`, no boxes, no ASCII rules. The license header is the sole exception.
- Never write -- inside a comment (the ASCII rule keeps the em dash out, and a double hyphen reads as a typo); rephrase instead.
- These rules govern the comments a change **writes**. Do not restyle existing comments along the way: one that already earned its size keeps it, and reflowing a file's comments buries the actual change exactly as a bulk reformat does.

```
/* one line is the default, and most comments need no more */

/**
 * A block earns its size by naming a trap: what was already gotten wrong once,
 * and would be gotten wrong again without the warning.
 */
```

Blank Lines

- Never three or more consecutive blank lines, anywhere.
- Exactly two blank lines separate function definitions in a `.c` or `.cl` file. **A header may use one**, and the files that carry small `LIBXS_API_INLINE` definitions are uniform about it: `libxs_gemm.h` and `libxs_token.h` use one throughout, `libxs_math.h` for sixteen of its twenty-one. One or two, but not a mixture within a file, which is the surrounding code a change there has to match.
- At most one blank line separates logical blocks inside a function. Two is what separates the functions themselves, so two never appear inside one.

Formatting

A `.clang-format` file is present (LLVM-based, 2-space indent, 96-column limit, never tabs), and `scripts/tool_clangformat.sh` selects the newest available version of the tool.

Do not bulk-reformat as part of a change. Recent clang-format versions reflow entire files, which buries a small change in hundreds of unrelated lines and makes review impossible. This is why clang-format is deliberately absent from the hook set. Format new and edited code by hand to match the file around it; the formatter is a maintenance tool, run deliberately and committed on its own.

Do not mix reformatting, renaming, and behavioural change in one commit.

Where the opening brace goes. It stays on the line of the construct it belongs to — `if (0 < n) {`, `for (...) {`, `} else {` — with two exceptions, and both are checked:

- **A function body opens on its own line in an implementation unit**, a `.c` or a `.cl` file: 1971 definitions do it and 78 do not.
- **A brace whose parentheses were broken across lines opens on its own line**, because the brace is then what tells the reader the condition has ended:

```
if (0 == first &&
    0 != second)
{
  ...
}
```

In a header the brace may trail the signature, and it usually should. What rules a header is a readable API: the declarations are read as a list, and an inline definition sits in that list. It is also outside the ABI and meant to stay a small tool, so a body long enough to want the brace on a line of its own needs a reason — `libxs_gemm.h` has one. Both forms are accepted there, which is the difference from an implementation unit; a broken parameter list still takes the brace onto its own line.

A bare block has no construct line to sit on, so nothing is required of it: `{ int scope_ = n;` and a brace alone on the line are both in use and both fine. A struct, a union and an initializer are left alone as well.

Whatever an `if`, an `else` or a loop controls stays on the keyword's line, or it is braced. Once the construct reaches a second line the braces are what say where it ends, and that includes the case where only the condition wrapped:

```
while (npos < (int)ctx->text_size
  && 0 != isspace(ctx->text[npos]))
{
  ++npos;
}
```

Each keyword is judged on its own, so an `if` with a braced body and a one-line `else` is two decisions rather than one. An `else if` is the inner `if` and is judged there.

Two more places keep the brace on its own line, and the check knows both. One is a construct whose line is followed by a preprocessor directive: the line above the brace is then `#endif`, and moving the brace up would carry it into the branch. The other is a header that is included inside a function body, where the file's own control flow sits at brace depth zero and is not a definition.

```
#if defined(LIBXS_CPUID_ARM_MODEL_FALLBACK)
  if (NULL != info)
#endif
  {
    ...
  }
```

A directive in that position also excuses the braces themselves: the keyword and what it controls then belong to different configurations, and one pair of braces cannot serve both.

Indentation is not reformatted by a hook, but one thing about it is checked: **two closing braces in a row must step left.** Sharing a column means they close blocks that are nested, so a level is missing from the indentation even though the braces balance and the compiler is content. The check is local and makes no assumption about the indent unit, and a preprocessor directive between the two resets it, because which brace belongs to which block then depends on the configuration. `} else {` is not a closer for this purpose: it reopens, so the next closer legitimately shares its column.

Library Code

- Library code does not terminate the process: no direct `exit(...)` in `src/`. Return a status and let the caller decide. The macro that wraps the one unavoidable case is the single exception.
- Environment variables carry the project's own prefix (`LIBXS_*` or `LIBXSTREAM_*`). `scripts/tool_checkenvars.sh --list` lists what the source reads, ours and foreign.
- **Header-only mode must keep working.** The amalgamated header (`*_source.h`, or the corresponding `-D*_SOURCE`) has to be includable from multiple translation units, so a new file-scope symbol in `src/*.c` needs internal linkage or the established macro treatment.
- Fix the cause, not the symptom. When a sample, test, or dependent project runs into a limitation of the library, change the library rather than working around it at the call site.

Fortran

Fortran sources are **fixed-form** (`.f`). Free-form (`.f90`) is not used; please do not propose converting them.

- Statements occupy columns 7 to 72. Columns 73 and beyond are ignored, so text reaching column 73 is **silently truncated** rather than diagnosed.
- A continued line carries `&` in column 6. Sources additionally place a trailing `&` in column 73, so the same file also reads as free-form; that marker is the only thing allowed at column 73.
- `.fi` files are included into fixed-form sources and follow the same rules.

Scripts

- POSIX-portable shell. `sed -i` is rejected: it is not portable (macOS).
- Shell scripts pass `shellcheck`, which the hooks run.
- Python is formatted with `black -l79` and passes `flake8`; `mypy` covers the tooling under `scripts/` and `.theme/`, not sample code.
- Where a hook carries an exclusion, it names the open findings it defers. An exclusion is a backlog item, not a permission.

Documentation

Documentation is terse and written for the person *using* the code, not for the person who wrote it.

- **Every `README.md` is user documentation**: what the thing does, how to enable it, how to run it, which environment variables and build knobs exist. Nothing more.
- Insights stay out. Design rationale, derivations, and performance analysis do not belong in a README — the single exception being a surprising usability implication the user has to know about (a knob that silently changes accuracy, a mode that only works on one vendor). If it does not change how someone uses the code, it is not user documentation.
- A short *why* belongs in the commit message.
- Document every new environment variable. A variable that `tool_checkenvars.sh` reports but the documentation does not mention is a defect, and the hook of the same name says so.
- A new page under `documentation/` needs a `nav` entry in `mkdocs.yml`.

Markdown may use any UTF-8, but the PDF is produced through LaTeX, which cannot render an arbitrary glyph. `PDF_UTF8_SED` in `Makefile.inc` transliterates a known set (Greek letters, arrows, comparison and set operators, ceiling and floor brackets). If `make documentation` fails on a character, add it there rather than removing it from the text.

Parts of `documentation/` are **generated**, and editing the output is lost work: the landing page comes from `README.md`, the development page from `scripts/README.md`, one page per sample from `samples/*/README.md`, one page per test from `tests/*.c`, this page from LIBXS, and the PDFs from all of the above. Version and amalgamated headers are generated too. Edit the source, then:

```
make documentation   # PDFs
make mkdocs          # serve the site with live reload
make mkslides        # serve a slide deck (SLIDES=<topic>)
```

The Makefile is authoritative about what is generated from what.

Build Systems

GNU Make is primary. The `Makefile` and the shared `Makefile.inc` define the defaults, the knobs, and the behaviour; they are the reference. **CMake is secondary**: it has to produce the same artifacts, but it does not get to define anything.

- A new source file, build knob, or changed default lands in the Makefile first, then in `CMakeLists.txt`. `CMakeLists.txt` mirrors the Makefile's defaults and says so where it matters — when a default moves, move both.
- Every step is taken to keep the two interchangeable *from the consumer's side*. In particular, `make` writes the files a CMake consumer expects — the package configuration under `lib/cmake/<project>/` alongside the pkg-config `.pc` files — so a tree built and installed with GNU Make is usable via `find_package()` without CMake ever having run. GNU Make pretending to be CMake is a feature, not a workaround.
- Continuous integration builds and installs both ways and then consumes the result via `find_package()` and pkg-config. A change that only works in one of the two is incomplete.

Building and Testing

Build without `DBG=1` by default. Debug builds are `-O0`, which makes any runtime-bearing sample prohibitively slow — never measure performance with one.

```
make -j $(nproc)                      # default build
make -j $(nproc) test                 # test suite
make -j $(nproc) DBG=1 PEDANTIC=2     # correctness check before submitting
```

`PEDANTIC=2` enables strict warnings and `ANALYZE=1` runs the compiler's static analyzer; where a project ships `scripts/tool_analyze.sh`, that runs cppcheck on top. A change is expected to be warning-free under `DBG=1 PEDANTIC=2`, because that is what continuous integration builds: GCC, Intel oneAPI, and macOS, covering release and strict debug configurations as well as a header-only build compiled as C++. See `.github/workflows/` for the exact matrix.

`SANITIZE=<kind>` passes `-fsanitize=<kind>` to both the compiler and the linker. ThreadSanitizer needs no options file: `__tsan_default_options` and `__tsan_default_suppressions` in `libxs_main.c` carry what a run of the library needs, and the environment is parsed after them, so `TSAN_OPTIONS` still overrides either one. They are inert in a shared-library build, where the runtime's own weak definition is found first, and that build needs the environment instead.

```
make SANITIZE=thread
setarch -R ./sample.x    # ASLR off, or the runtime aborts before main()
```

Some kernels abort with `unexpected memory mapping` unless ASLR is off, hence the `setarch`. Exit code 66 means races were reported, not that the program failed.

GNU libgomp is not instrumented, so a race whose two accesses sit on opposite sides of an `omp parallel` region is reported even though the region's own barrier separates them; a fifteen-line correct program reproduces it. The compiled-in `ignore_noninstrumented_modules=1` cuts the volume about fivefold but not the class. **Triage by region position, not by thread**: a report is that artifact unless *both* stacks carry an `_omp_fn` frame. Thread identity is not the test, since the main thread is a member of the team and takes part in the rendezvous. Suppressing the class is not possible, because those stacks run through the code under test; building with Clang against a libomp that has TSan support (it ships `libarcher.so`) is what makes the region boundaries visible.

ABI and Versioning

The version derives from Git tags and `version.txt`; the shared library carries an SOVERSION. `scripts/tool_checkabi.sh` compares exported symbols against the recorded baseline. Do not remove or rename a published symbol — add a new one instead. Run the checker on a build that has symbol information:

```
make STATIC=0 SYM=1
scripts/tool_checkabi.sh
```

A symbol name outside the project's namespace is an error, not a warning.

Commits and Pull Requests

- One concern per commit. Keep unrelated formatting out of it.
- Subject line: short, capitalized, no trailing period, e.g. `Improved device memory allocation`. An area prefix is fine: `CMake: updated to test kernels`.
- Explain *why* in the body when the subject cannot carry it.
- Contributions are accepted under the project's BSD-3-Clause license.

Before submitting:

```
scripts/tool_normalize.sh --install       # once per clone, then automatic
scripts/tool_normalize.sh                 # whitespace, encoding, lint
make -j $(nproc) DBG=1 PEDANTIC=2 test    # strict build and tests
scripts/tool_checkabi.sh                  # only if public symbols changed
```

Workflow with Assistants

If an AI assistant is used, the same policies apply, plus two more:

- Discuss design options and trade-offs before implementing.
- Present options as inline text, not as interactive choice widgets.